



Decision support system for Forest fire management using Ontology with Big Data and LLMs

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Abstract

Forests are crucial for ecological balance, but wildfires are a major cause of forest loss, posing significant risks. Fire weather indices, which assess wildfire risk and predict resource demands, are vital. With the rise of sensor networks in fields like healthcare and environmental monitoring, semantic sensor networks are increasingly used to gather climatic data such as wind speed, temperature, and humidity. However, processing these data streams to determine fire weather indices presents challenges, underscoring the growing importance of effective forest fire detection. This paper discusses using Apache Spark for early forest fire detection, enhancing fire risk prediction with meteorological and geographical data. Building on our previous development of Semantic Sensor Network (SSN) ontologies and Semantic Web Rules Language (SWRL) for managing forest fires in Monesterial Natural Park, we expanded SWRL to improve a Decision Support System (DSS) using a Large Language Models (LLMs) and Spark framework. We implemented real-time alerts with Spark streaming, tailored to various fire scenarios, and validated our approach using ontology metrics, query-based evaluations, LLMs score precision, F1 score, and recall measures.

Keywords LLMs · Spark · SSN · DSS · Forest Fire

Abbreviations

SSN	Semantic Sensor Network	CSV	Comma Separated Value
DSS	Decision Support System	FAISS	Facebook AI Similarity Search
LLMs	Large Language Models	SVM	Support Vector Machine
OWL	Ontology Web Language	DOLCE	Descriptive Ontology for Linguistic and Cognitive Engineering
RDF	Resource Description Framework	YOLO	You Only Look Once
CFWI	Canadian Fire Weather Index	EE-SS	Energy Efficient Clustering Algorithm
BUI	Buildup Index	UAV	Unmanned Aerial Vehicle
FFMC	Fine Fuel Moisture Code	RH	Relative Humidity
SWRL	Semantic Web Rule Language	JSON-LD	JavaScript Object Notation for Linked Data
FWI	Fire Weather Index	XML	Extensible Markup Language
DMC	Duff Moisture Code	IoT	Internet of Things
DC	Drought Code		
ISI	Initial Spread Index		

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1 Introduction

Forest fires constitute a crucial worldwide undertaking [1], posing giant threats to each ecological integrity and monetary balance. Annually, these fires devour thousands and thousands of hectares of forests worldwide, leading to the destruction of essential habitats, a decline in biodiversity, and the release of good-sized amounts of carbon emissions into the environment, further hectic the influences of

climate change. While herbal phenomena like lightning can provoke forest fires in particular situations, the major causes are human activities, consisting of agricultural burning, land clearing, and accidental ignition.

India stands out as specifically vulnerable to forest fires [2] due to many things, including big human encroachment near forested regions, unsustainable land-use practices, and inadequate fire management strategies. Research suggests that a considerable majority of fires in India are anthropogenic in the beginning rather than being indeed precipitated. These fires now devastate woodland ecosystems and endangered wildlife and directly harm human communities dwelling in or near woodland areas. Moreover, these fires' smoke and particulate matter contribute considerably to air pollutant levels, compromising public health and environmental excellence.

The escalating influences of climate exchange exacerbate the problem by extending the period and intensifying the severity of heart seasons globally. Consequently, there is a pressing need to develop and implement more powerful and localized fire management and prevention strategies. Most forest management companies rely closely on fire-place threat score systems to assess health hazards and allocate firefighting sources. However, a number of the present worldwide fireplace chance score systems, such as the Canadian Fire Weather Index (CFWI) [3] [4] and the McArthur Forest Fire Danger Index [5] [6], were advanced based on weather and ecological facts from areas with one of a kind climatic and ecological traits. As a result, these systems might not reflect the unique and diverse conditions universal in Indian forests.

Spark is a unified analytics engine designed to efficiently perform several operations on big data. It can integrate easily with LLMs to design an efficient real-time DSS. This integration is beneficial when working with vast amounts of data to interpret them and make automated reasoning more accurate [7]. Working with Spark and LLMs enables us to make accurate decisions based on prespecified rules and environments for real-time operations. LLMs can perform automated reasoning on data processed through Spark and can integrate the capability to enhance the logic and accuracy of rule-based queries, particularly for complex decision-making tasks.

Ontologies are employed to establish a unified data model within the information space, streamlining several critical aspects: (1) They provide comprehensive details on diverse data sources and infrastructure services, incorporating domain-specific insights; (2) They create a common lexicon that enhances the infrastructure's capability to interact with external sources that utilize various methodologies; and (3) They improve the processes of discovering, accessing, and integrating data. These ontologies cater to a wide range of knowledge representation requirements:

(1) The SSN [8] Ontology specifically models sensor networks and captures data concerning distinct features of interest. DOLCE + DnS UltraLite [9] serves as the upper ontology in this framework; (2) It showcases the resources provided by the infrastructure and the accessible datasets.

This research work addresses several challenges:

1. Real-time data processing with Spark enables handling large volumes of sensor data to support quick decisions. Integration with LLMs ensures that varying data volumes do not affect the model's performance or uptime.
2. It identifies suitable data sources based on their content, which includes features, observations, sensors, and dataset-defined scope.
3. It tackles the heterogeneity of data sources, including different data models, interfaces, and mobility issues.
4. It involves users in data collection based on domain concepts, synthesizing information from various sources to enhance problem understanding.

The proposed approach utilizes semantic technologies in decision support systems for forest fires to manage diverse data sources and facilitate data correlation. This enables user-level applications to create ontology-based queries, which can then be transformed into executable queries for data sources. Spark streaming with LLMs allows the system to interact with textual data in real time and make effective decisions based on a predefined set of rules and queries.

The system is designed to achieve the following objectives and meet specific user needs:

1. Develop a SWRL for a rule-based DSS that can be represented using classes, properties, and individuals.
2. To present a Real-time LLMs-based alert generation model.
3. To establish an index to showcase fire weather threat ratings, formulated with meteorological experts' guidance, using Ontology Web Language (OWL).
4. To effectively manage and process large-scale sensor data.
5. We transform raw data into Resource Description Framework (RDF) triples to enhance data quality.

The rest of the section of this research work is organized as follows: Section II presents the background and related work to identify the research gap. Section III expounds on the suggested research methodology. Section IV presents the experimental works and their results. Section V discusses the conclusions and recommendations for future work.

2 Literature review

In this section, we will discuss the related works done in recent years and identify the research gaps that need further exploration using the underlying methodology shown in Table 1. Recent advancements in forest fire detection have seen a variety of innovative approaches leveraging deep learning, IoT, and wireless technologies.

Abdusalomov et al. [10] developed an automated method using an improved Detectron platform, achieving a remarkable precision of 99.3% in detecting fires under various conditions, both day and night. This method utilized a custom dataset and significantly improved fire detection accuracy.

Burak et al. [11] proposed a hierarchical forest fire detection system using a combination of multimedia and scalar sensors in IoT systems, achieving a validation accuracy of 98.28% while reducing energy consumption by 29.94%. Avazov et al. [12] introduced a fire detection method tailored for smart cities, leveraging an enhanced YOLOv4 [13] based convolutional neural network capable of triggering an alarm within eight seconds of a fire outbreak with high speed and accuracy.

On a different front, Karim et al. [14] used Wireless Sensor Networks (WSNs) with mid-point K-means clustering to forecast high-activity forest zones with 98% accuracy, ensuring energy-efficient data transmission. Bharany et al. [15] focused on using Flying Ad Hoc Networks (FANETs) for fire detection, employing the EE-SS clustering algorithm, a clustering methods that optimize energy consumption while ensuring scalability in sensor networks which extended UAV lifetime and outperformed other state-of-the-art algorithms. These studies demonstrate the growing integration of advanced technologies such as deep learning, IoT, and WSNs in enhancing forest fire detection systems' accuracy, efficiency, and speed.

The literature review emphasizes the necessity of an effective model that integrates Spark streaming, LLMs, and ontology-based methods. This integrated approach aims to lower costs and achieve greater accuracy than current systems such as satellite-based, optical camera-based, and human-based forest fire observation systems. While various ontology-based decision support systems for forest fires exist, such as the BACAREX ontology, Geo-ontology, and FWI ontology, they primarily concentrate on observation, operational tracking, and FWI calculation. None of the reviewed studies address an integrated approach that merges LLMs and the Spark framework within a single ontology.

This research aims to create a framework within the SSN ontology that allows the ability to search observation systems using SPARQL and improves interactivity for

computer programs and human users alike. By improving the understanding and application of forest fire and surveillance data, this all-encompassing framework seeks to increase the information's comprehensiveness. With the help of LLMs, it uses Spark streaming to strengthen the framework and enable quick and efficient decision-making. The structure also illustrates the key components for controlling and monitoring forest fires.

3 Methodology

This section elaborates Ontology + LLMs + Spark Streaming framework, which generates real-time alerts by streaming static data through Apache Spark, enabling continuous processing. The system is based on the Montesinho Forest Fire Dataset, which includes spatial (X, Y coordinates), temporal (month, day), meteorological (temperature, humidity, wind speed, rainfall), and fire behavior parameters (Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC), Drought Code (DC), Initial Spread Index (ISI), and burned area). Although the dataset is static, Apache Spark streams it in batches, simulating a real-time data pipeline. Each batch is timestamped to maintain temporal order, ensuring a structured data flow. SWRL rules analyze fire risk factors dynamically, triggering alerts when predefined thresholds, such as high ISI, low humidity, or extreme temperatures, are met. LLMs process these alerts to provide precautionary suggestions based on fire conditions. Spark's distributed processing ensures the system remains scalable and efficient, handling large datasets effectively. While the model currently operates on historical data, it is adaptable for future integration with real-time IoT sensors, making it suitable for large-scale fire risk management.

3.1 Architecture of the proposed model

Figure 1 illustrates the comprehensive working model, showcasing the proposed combination of Spark streaming, ontology, and LLMs. It demonstrates how SPARQL queries are applied to both LLMs and ontology, facilitating decision-making for forest fire management based on the results obtained from this integration.

3.2 Dataset description

The dataset encompasses a detailed array of spatial, temporal, and meteorological data crucial for analyzing and predicting forest fire behavior in Montesinho Park [30, 31], Portugal. It includes precise spatial coordinates to pinpoint high-risk zones, temporal details that highlight seasonal and weekly trends in fire occurrences, and meteorological

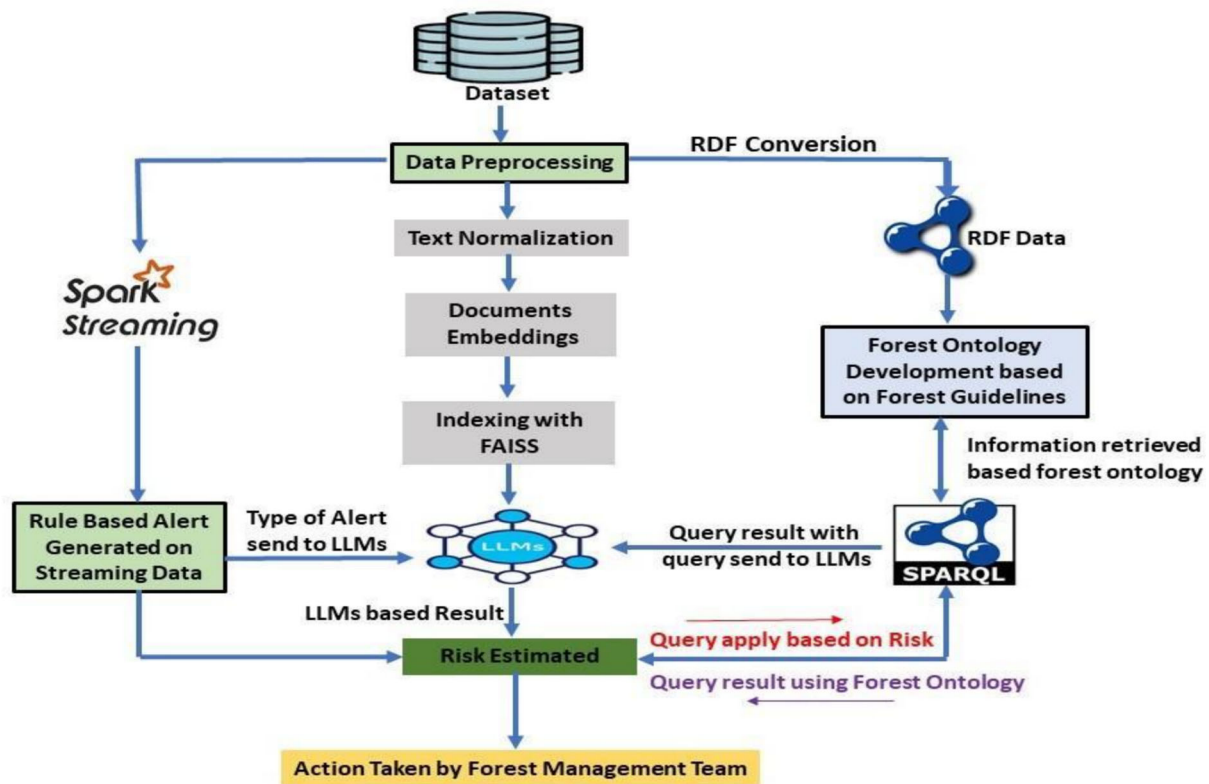


Fig. 1 Framework view of the complete working model

variables such as temperature, humidity, wind speed, and rainfall that directly affect fire dynamics. Furthermore, the dataset features a target variable, the burned area, which quantifies the extent of forest damage, providing valuable insights into fire severity. This variable, prone to skewness towards smaller values, is recommended to be transformed logarithmically for more accurate modeling. Overall, this dataset is instrumental for devising effective forest fire management and intervention strategies.

In Fig. 2, the matrix, we can interpret that temperature, ISI, DC, DMC, and FPMC are strongly correlated with each other, and each of these features is positively correlated to the area of fire burned as well, whereas RH Wind and rain are slightly positively correlated with each other and are negatively correlated to the area of the forest fire spread. This means that factors like RH, wind, and rain increase the area of the forest fire, and factors, like temperature, ISI, DC, DMC, and FPMC combined can cause more fire spread.

3.2.1 Data preprocessing

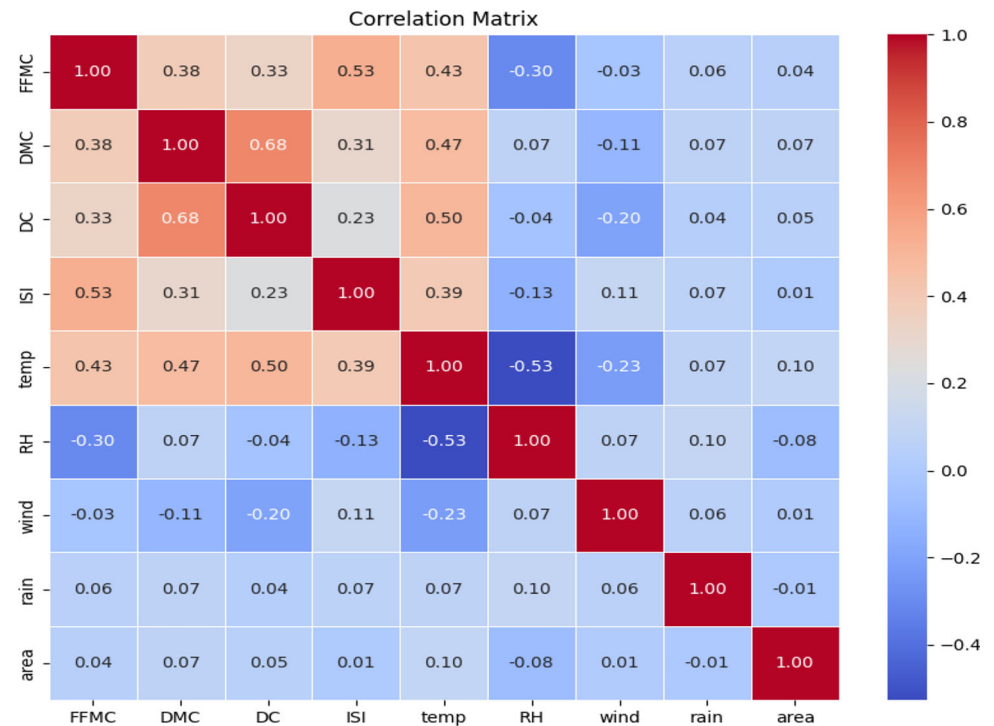
The data gathering and setup process begins with data streaming using PySpark (python library for apache spark) and importing necessary libraries, then creating a stream context to convert real-time data into a continuous

DataFrame stream for analysis. Data cleaning includes a logarithmic transformation of the skewed 'area' target variable to normalize its distribution for better modeling suitability. Numerical features like FPMC, DMC, DC, ISI, temperature, RH, wind, and rain are standardized using Z-score normalization to ensure zero mean and unit variance, enhancing model stability and convergence speed. Categorical variables 'month' and 'day' undergo one-hot encoding to transform them into binary columns, facilitating their inclusion in models requiring numerical inputs.

Feature selection is conducted through correlation analysis to exclude low-correlated or redundant features, improving model efficiency. Outlier detection and removal are managed with statistical methods like z-scores or the interquartile range, assessing their impact on model performance and modifying them accordingly; for the imbalanced data of the 'area' variable, techniques such as oversampling and undersampling are employed to balance class distribution, enhancing the model's capability to capture patterns effectively.

3.2.2 RDF conversion

RDF, developed in partnership with the World Wide Web Consortium (W3C), standardizes the description of online resources and facilitates seamless data transmission and

Fig. 2 Correlation matrix of the dataset

integration. By expressing data as subject-predicate-object triples, RDF defines clear relationships between data elements, enabling a flexible, structured approach to data management. One of RDF's key strengths is its ability to separate data from its structure, allowing for the integration of diverse data sources under a unified schema. In this research, we employ the python language to transform a CSV dataset into RDF format, thereby facilitating data querying and linking across various sources. The CSV format, while simple to work with for structured data, lacks the semantic richness needed for more complex data

modeling. Figure 3 shows the conversion into RDF data from CSV sources.

By converting the CSV data to RDF, we can enrich the dataset with metadata, making it more interoperable and accessible for linked data applications. Depending on the use case, we can serialize the RDF output from this conversion process in multiple formats like Turtle, XML, or JSON-LD.

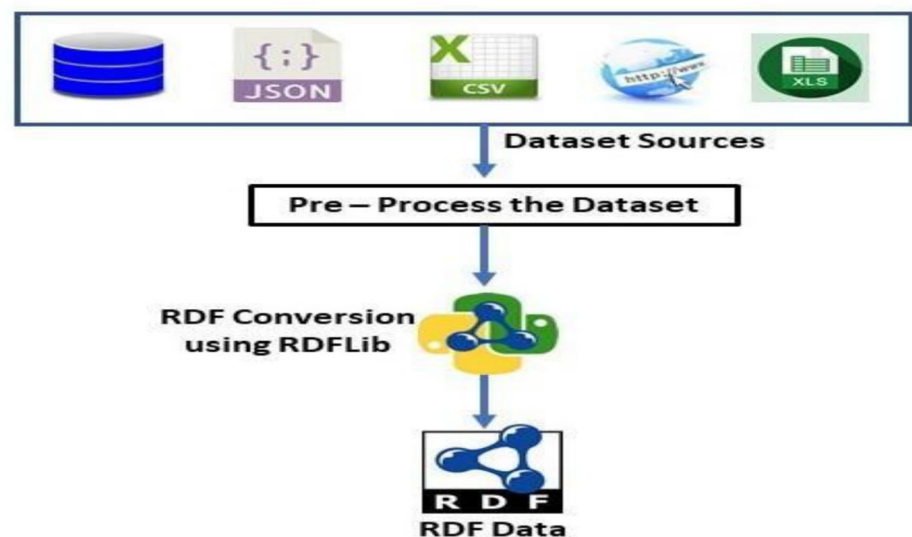
Fig. 3 Converting datasets to RDF

Table 1 Summary of existing works

Author-years	Objectives	Proposed methodology	Results	Tools and languages
Ginkal et al. 2024 [16]	Cloud-based wireless sensor network technology for forest fire detection	WSN, Computer Vision, UAV, YOLO, IoT, MODIS	Real-time monitoring improved efficiency over traditional methods	WSN, Computer Vision, UAV, YOLO, IoT, MODIS
Ibraheem et al. 2024 [17]	Early detection system using Intermediate Fusion VGG16 model and ECP-LEACH	Intermediate Fusion VGG16, ECP-LEACH	99.86% detection accuracy, regulated energy consumption	Python, Intermediate Fusion VGG16, ECP-LEACH
Huiyi et al. 2024 [18]	Identifying causes of forest fire events	JenK-DBSCAN model, MODIS MCD64A1 product, RF, LR, SVM	High accuracy in identifying fire causes, meteorological variables as main factors	JenK-DBSCAN, MODIS, RF, LR, SVM
Cumbane et al. 2019 [19]	Big data framework for disaster response application	Review various frameworks and how efficiently it can be used to solve natural disasters	Evaluating each framework on various parameters such as scalability, responsiveness, data processing, etc	Apache Spark, Kafka, and Hadoop
Athanasis et al. 2019 [20]	Big Data Analysis for Wildfire Prevention and Management	Role of Geospatial big data for wildfire management	contribute to cutting-edge knowledge sharing between wildfire conflict operation centers and field fire fighting units	Big data clusters
Rajasekaran et al. 2015 [21]	Big data analysis for forest fire management	Role of framework to real time alert generation	Thirty minutes before forest fire prediction to reduce the casualties	Apache Hadoop
Martins et al. 2022 [22]	To develop a forest fire susceptibility mapping approach using machine learning in Southeast China	Employed space-humidity-constrained (SHC-based) sampling method and dynamic factor extraction method based on DTW and precipitation thresholds (Thep)	RF model demonstrated the best performance with accuracy enhancements of over 8%	Random Forest (RF), Support Vector Machine (SVM), multi-layer perceptron (MLP)
Singh et al. 2024 [23]	To examine forest fire trends, patterns, and spatiotemporal distribution in tropical seasonal forests in Odisha State, India	Utilized MODIS imagery, Kernel density tools, and machine learning algorithms (SVM and RF)	Random Forest achieved over 94% precision in identifying forest fire susceptibility zones	SVM, Random Forest RF, MODIS imagery
Zhao et al. 2024 [24]	To map forest fire sensitivity in Hunan Province using machine learning models	Constructed four machine learning models: RF, SVM, XGboost, and BRT	FR-BRT model performed the best with an AUC of 0.97	RF (SVM), eXtreme Gradient Boosting (XGboost), Boosted Regression Tree (BRT)
Mishra et al. 2024 [25]	To develop a dual-model deep learning approach for super real-time wildfire spread forecasting	Utilized U-Net for prediction and ConvLSTM layers for real-time refinement	Over 90% agreement between AI forecasts and numerical simulations	U-Net, ConvLSTM, deep learning
Zhang et al. 2023 [26]	To predict post-fire burn severity in California using a novel machine-learning model	Employed the Super Learner (SL) algorithm accounting for spatial autocorrelation	SL model predicted burn severity with reasonable classification accuracy	SL, algorithm, Vecchia's Gaussian approximation
Li et al. 2024 [27]	To identify and prioritize forest fire-prone areas in Iran's Firouzabad region	Utilized GIS-based Bayesian and Random Forest methodologies	RF outperformed the Bayesian model with an AUC score of 0.876	RF, Bayesian model, GIS

Table 1 (continued)

Author-years	Objectives	Proposed methodology	Results	Tools and languages
Simafranca et al. 2024[28]	It introduces an updated version of the ONTO-SAFE ontology with improved semantic capabilities, effectively capturing a range of forest fire scenarios	Used Semantic Web technologies for developing ontology	Based on metrics count got the result of Average population 70.089552 Class richness 0.044776	SPARQL, RDF4j
Noroozi, et. 2024 [29]	Fine-tuned the LLMs for ontology construction due to limited domain expertise	Utilizes prompt engineering	Three ontology learning tasks were evaluated using fine-tuned LLMs	LLaMA-7B.BLOOM-3B

3.3 Ontology development

For the development of the forest ontology, we utilized the upper-level SSN ontology, which has a standardized forest ontology. The SSN ontology was developed by the Semantic Sensor Networks Incubator Group of the W3C. In our previous research, we based our ontology development on documents related to Canadian forest fires. However, in this current study, we have refined the forest ontology by incorporating documentation from other countries [2, 32–35], like India, to increase its accuracy and to encompass a broader scope of knowledge regarding forest vegetation.

We have enhanced the ontology by incorporating a diverse array of policies, plans, and forest fire prevention rules, together with coordination strategies across agencies, as detailed in forest documentation from various countries. This improvement broadens the ontology's scope and applicability to forest fire management. Additionally, we have integrated numerous SWRL rules into the forest ontology, drawing on documentation from different nations to ensure a more robust and versatile framework [36].

3.3.1 SWRL rules development

The SWRL [37] merges the OWL with the Rule Markup Language, offering a high-level abstract structure for rules similar to Horn clauses. This model's rules are formulated using OWL concepts such as classes, properties, and individuals. In our previous work, we incorporated 42 rules; however, in this current study, we have expanded the number to 110. Additionally, we have developed more complex rules based on our existing findings, details of which are presented in the following sections. These rule-based transformations are applied to streaming data.

Previous work established rules that depend on specific columns such as FFMC, DMC, DC, ISI, rain, and wind. Various classes for each column value (e.g., extremely easy, difficult, fast) have been categorized using predefined thresholds. In this extended version, we enhance the work by covering all different scenarios, such as human resource management in emergencies and determining which type of firefighting equipment is needed at which time. This makes the model concerning forest management more complete. Some rules are shown in Table 2, Table 3, and Table 4. We have accessed several documents [38] to develop these rules [39] [40].

3.4 Spark streaming

Spark Streaming is the core of the Apache Spark [41] API, providing scalable solutions for high throughput and fault

Table 2 SWRL rules for the elaboration of a fire management plan

S.No	SWRL rules	Explanation
1	PreventiveAction(?action)^hasScenario(?action,?scenario)^hasIgnitionRisk(?scenario,?risk) ^swrlb:lessThanOrEqual(?risk,0.5)- > reduceIgnitionRisk(?action)	When the ignition risk of a scenario drops below a specified threshold (e.g., 0.5), indicating low risk, a rule is triggered. This rule recommends a preventive action to lower the ignition risk further
2	PreventiveAction(?action)^hasScenario(?action,?scenario)^hasBurnedArea(?scenario,?area) ^swrlb:lessThanOrEqual(?area,1000)- > limitBurnedArea(?action)	This rule suggests a preventive action to limit the burned area if the projected burned area for a scenario is below a certain threshold (e.g., 1000 hectares)
3	PreventiveAction(?action)^hasScenario(?action,?scenario)^hasDamage(?scenario,?damage) ^swrlb:lessThanOrEqual(?damage,20)- > limitDamage(?action)	If the estimated damage for a scenario is below a specified threshold (e.g., 20%), this rule recommends an action to limit the damage caused by fire
5	PreventiveAction(?action)^infrastructureInstallation(?action)- > installInfrastructure(?action)	This rule suggests installing preventive infrastructure such as lookout towers, tracks, water points, or fuel breaks
6	PreventiveAction(?action)^populationSensitization(?action)— > sensitizePopulation(?action)	This rule recommends sensitizing the population and involving them in preventive activities in the forest
7	PreventiveAction(?action)^forestStandIntervention(?action)- > nterveneForestStand(?action)	This rule suggests interventions in forest stands, such as undergrowth clearing and other forest operations

Table 3 SWRL rules for human resources management in fire suppression

S. No	SWRL rules	Explanation
1	SpecializedPersonnel(?person)^hasTraining(?person,“Firefighting”)- > deploySpecializedPersonnel(?person)	This rule suggests deploying personnel with specific firefighting training to handle forest fire suppression operations
2	NotSpecializedPersonnel(?person)^hasInvolvement(?person,“Limited”)- > limitInvolvement(?person)	This rule recommends limiting the involvement of personnel without specialized firefighting training due to safety concerns and the potential for ineffective firefighting
3	Firefighter(?person)^hasIndividualProtectiveEquipment(?person, “Protective Gear”)—> ensure safety(?person)	Under this rule, firefighters are equipped with appropriate protective gear to ensure their safety during firefighting activities
4	AdministrativeAuthority(?authority)^hasResponsibilityModel(?authority,“IntegratedForester”)- > delegateResponsibility(?authority,“ForestManagementServices”)	This rule specifies the delegation of responsibility for forest fire suppression to forest management services under the “Integrated Forester” model
5	FireSuppressionService(?service)^hasClearOrganization(?service,true)- > ensureCoordination(?service)	This rule emphasizes the importance of clear organization within the fire suppression service to ensure effective coordination during firefighting operations
6	Zone(?zone) ^ hasRiskLevel(?zone, “High”)— > deployOptimalDensity(?zone,“OneBrigadePer5000Ha”)	This rule suggests deploying an optimal density of firefighting personnel, such as one brigade per 5000 hectares, in zones with high fire risk

Table 4 SWRL rule for firefighting equipment selection criteria

S. No	SWRL rules	Explanation
1	FireSuppressionOperation(?operation)^hasFireType(?operation,“Surface”)- > deployTerrestrialEquipment(?operation,“StandardEquipment”)	This rule suggests deploying terrestrial equipment, specifically standard equipment like shovels, hoe-rakes, and fire beaters, when the fire type is surface fire. This equipment is effective for initial attacks and low-intensity fires
2	FireSuppressionOperation(?operation)^hasDevelopmentPhase(?operation,“LargeUncontrolledFire”)— > useInitialAttackVehicles(?operation)	When facing large, uncontrolled fires, this rule recommends utilizing initial attack vehicles equipped with water tanks. These vehicles provide immediate response capabilities and can suppress or slow down fires until reinforcements arrive
3	WaterTanker(?vehicle)^hasWaterCapacity(?vehicle, ?capacity) ^ swrlb:lessThan(?capacity, 5000)— > deployMultipleVehicles(?vehicle)	For water tankers with capacities less than 5000 L, it suggests deploying multiple vehicles instead of a single large one. This ensures flexibility and mobility while maintaining sufficient water supply for firefighting operations
4	FireSuppressionOperation(?operation)^requiresIndirectIntervention(?operation,true)- > deployBulldozers(?operation)	In cases where indirect intervention occurs, such as building fire lines, this rule recommends deploying bulldozers. These machines can clear vegetation to create fire breaks, limiting the spread of the fire
5	FireSuppressionOperation(?operation)^hasEnvironmentalConditions(?operation,“DifficultAccess”)— > useHelicopters(?operation)	In areas with difficult access, such as rugged terrain, this rule suggests using helicopters for aerial firefighting operations. Helicopters are more maneuverable and can access remote areas where fixed-wing aircraft might face challenges
6	FireSuppressionOperation(?operation)^hasFireBehavior(?operation,“RapidSpread”)- > useShortTermRetardants(?operation)	When dealing with fires exhibiting rapid spread, this rule recommends using short-term retardants as chemical additives. These retardants increase water retention in vegetation, helping to slow down the fire’s advance

**Fig. 4** Stream processing through spark framework [42]

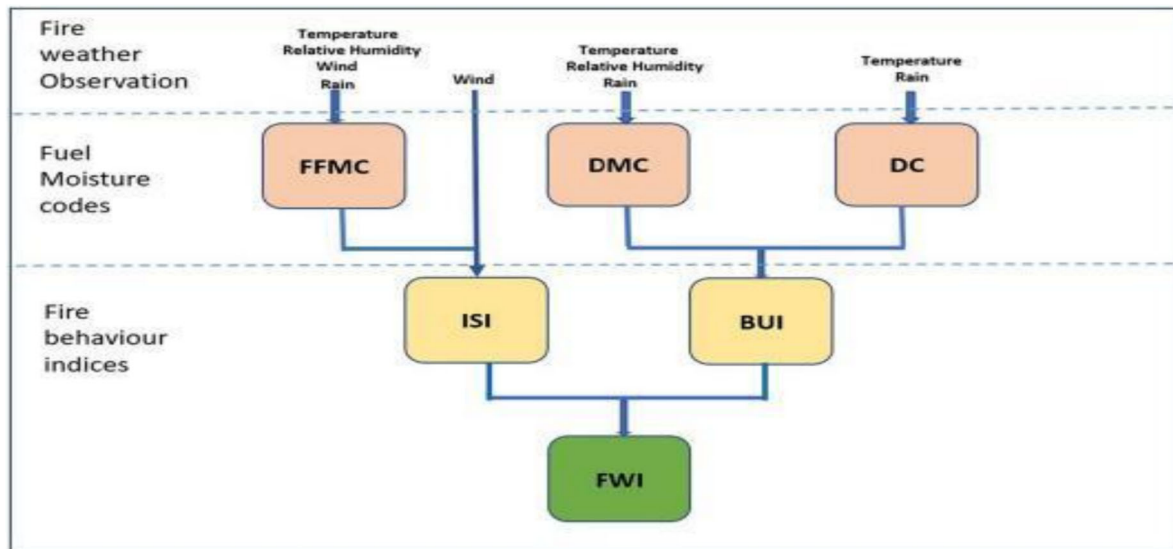


Fig. 5 Flowchart of the fire weather index [43]

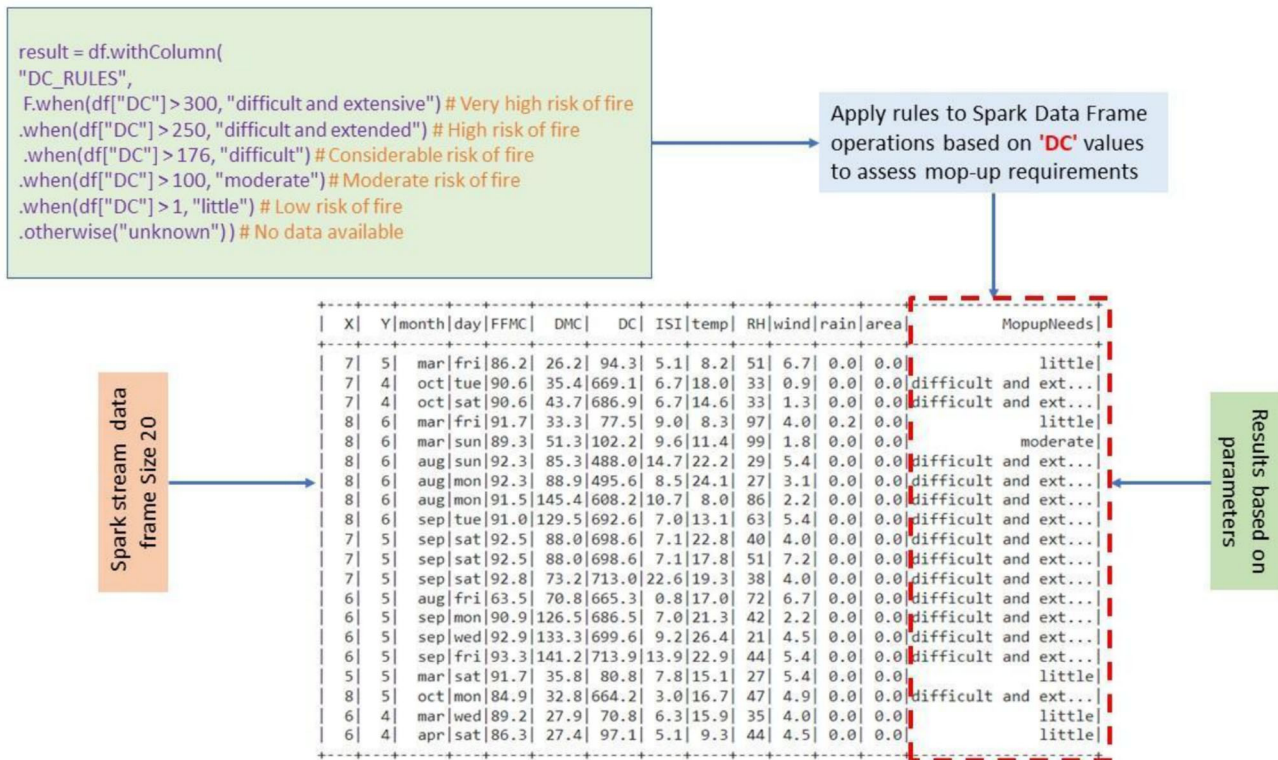


Fig. 6 Rules applied on DC values on stream data and tell about mop-up Needs

tolerance in processing live data streams. It supports both real-time and batch processing to maintain a low-latency approach suitable for various applications. Spark Streaming handles real-time data by dividing it into small, manageable batches (micro-batches), allowing near-instantaneous analysis and response to incoming data. For example, in forest fire monitoring systems, sensors

deployed in remote areas continuously generate environmental data such as temperature, humidity, and smoke levels. Spark Streaming ingests this sensor data in real-time, processes it immediately upon data generation, and triggers alerts if specific thresholds are crossed, helping authorities respond quickly to fire outbreaks. The platform enables replication of data across multiple nodes,

```

model_name = "Intel/dynamic_tinybert"

tokenizer = AutoTokenizer.from_pretrained(model_name, padding=True, truncation=True, max_length=512)

question_answerer = pipeline(
    "question-answering",
    model=model_name,
    tokenizer=tokenizer,
    return_tensors='pt',
)

llm = HuggingFacePipeline(pipeline=question_answerer, model_kwargs={"temperature": 0.7, "max_length": 512},)

```

Fig. 7 Model used for LLM Intel/dynamic_tinybert

enhancing robustness and ensuring data availability even in the event of network or node failures. Figure 4 shows Spark Streaming processing data in batches.

3.4.1 Spark streaming based alerts

To detect events and generate alerts on streaming data, we implement rules based on the FWI. The development of these rules incorporates parameters such as the Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC), Drought Code (DC), Initial Spread Index (ISI), Buildup Index (BUI), and FWI, derived from prior research. Figure 5 illustrates the FWI process through a detailed flow chart.

Figure 6 provides detailed information on how rules are applied to streaming data. DC rules are applied to batches of size 20, yielding results based on specific parameters. These results, categorized as “Mop up Needs”, encompass various parameters as depicted in Fig. 6.

DC elucidates the necessity of mop-up operations, highlighting the degree of controllable fire risk and facilitating appropriate action. The forest ontology contains all of the forest’s location details, making it easy to take action in areas with different types of forests, whether dense or less dense. We generate other FFMC, ISI, BUI, DMC, and FWI alerts based on stream data in the same manner.

3.4.2 Implement an alert system that utilizes LLMs for processing queries based on Spark alerts

The research harnesses advanced techniques to manage and interpret data related to forest fires, transforming this data into a refined text format. A thorough preprocessing phase initiates the process, cleansing the data of discrepancies and noise to ensure reliability and quality. Advanced text normalization methods normalize the clean data to achieve uniformity across the dataset, preparing it for further analytical processes.

We then convert the normalized text into document embeddings using the state-of-the-art all-MiniLM-L6-v2

[44] model. These embeddings are essential, as they capture the deep semantic meaning of the text, enabling accurate retrieval of relevant information. This transformation ensures that both significant and minor information is preserved in a searchable format. To support fast and efficient retrieval, we use FAISS [45] to build an indexing system over the embeddings. This enables scalable and rapid access to semantically similar documents during the querying phase.

At the core of the system is a retrieval-based question-answering (QA) framework powered by the Intel/dynamic_tinybert model [46] shown in Fig. 7, a transformer-based QA model available on Hugging Face. As an extractive QA model, it processes structured input as a question-context pair, where the context is retrieved using FAISS. The model then extracts the most relevant answer span from the provided context. Since this approach does not involve open-ended prompts or in-context learning, no prompt engineering was necessary. Additionally, no fine-tuning was performed; the system uses the pre-trained weights directly from Hugging Face to ensure reproducibility and facilitate real-world deployment.

To enhance contextual continuity during interactions, we incorporate a conversation buffer memory, allowing the system to maintain awareness across multiple queries. The system’s real-time capabilities further enable it to handle immediate questions related to environmental conditions, sensor outputs, or fire alerts with speed and precision. Finally, integration with Apache Spark enables distributed processing, ensuring that the system scales effectively to handle large datasets and multiple simultaneous queries without performance degradation.

The deployment of pre-trained LLMs alongside Apache Spark boosts the system’s speed and efficiency and its adaptability and scalability. This ensures that as fire safety requirements evolve alongside advancements in sensor technology and changes in environmental conditions, the system can adapt and continue to provide reliable, cost-effective fire alarm trigger detection, thus safeguarding

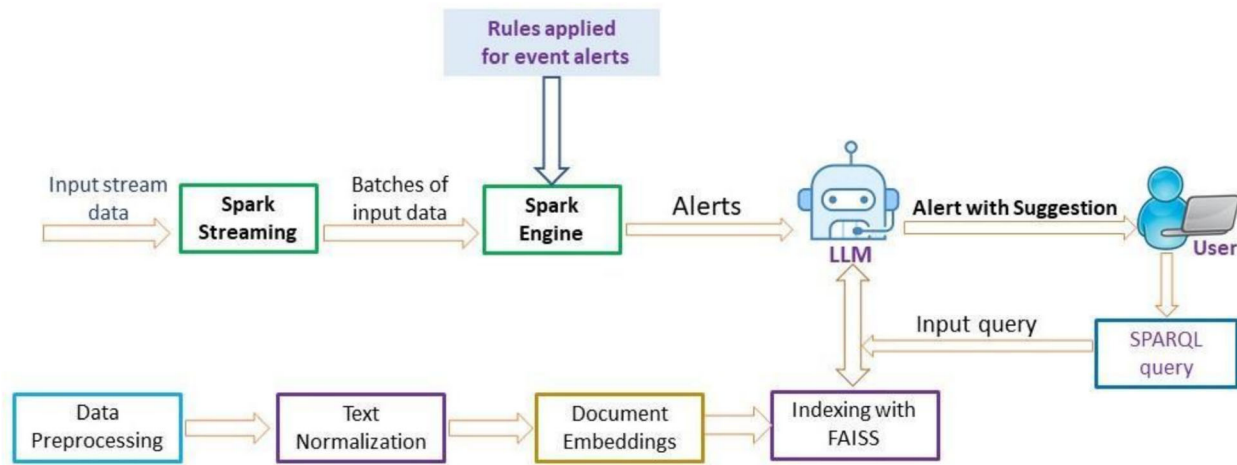


Fig. 8 Integration of spark framework with LLMs

Table 5 SPARQL query and corresponding result

	Query	Result
SPARQL query ontology	SELECT ?region ?temperature ?humidity ?wind	region temperature humidity wind "Pine Valley" 35 25 34
	WHERE { ?region rdf:type ex:ForestArea ?region ex:hasTemperature ?temperature ?region ex:hasHumidity ?humidity ?region ex:hasWindSpeed ?wind FILTER (?temperature > 30 && ?humidity < 30 && ?wind > 30) }	"Oak Ridge" 34 20 32
SQL query on LLMs	SELECT region, temperature, humidity, wind FROM ForestAreas WHERE temperature > 30 AND humidity < 30 AND wind > 30;	Predictive Insight: Based on the current dry spell and wind conditions, South Forest, although currently at 28 degrees and 32% humidity, will likely exceed high-risk thresholds within the next 48 h Precautionary measures are recommended: Resource Allocation Suggestion: Due to the severe risk parameters observed, immediate fire watch alerts and additional surveillance drones are suggested for North Forest

infrastructure and lives with heightened efficacy, as shown in Fig. 8.

3.5 SPARQL query-based LLMs alert and ontology alert generation

When the same SPARQL query is executed, the results from both the LLMs-based alert system and the ontology-based alert system are nearly identical. However, leveraging its advanced learning capabilities, the LLM system frequently provides additional insights and suggestions, which are shown in Table 5. Subsequently, the ontology system evaluates these results from the LLMs, employing

its extensive and detailed knowledge base. This base includes critical information such as location specifics, temperature readings, and data pertaining to sensor deployment areas, thereby ensuring that the LLMs suggestions are both relevant and accurate [43].

Figure 9 shows how we ask questions to both the ontology and LLMs. It demonstrates how LLMs find information using a system called FAISS. This helps us understand how both methods work together to get the necessary information.

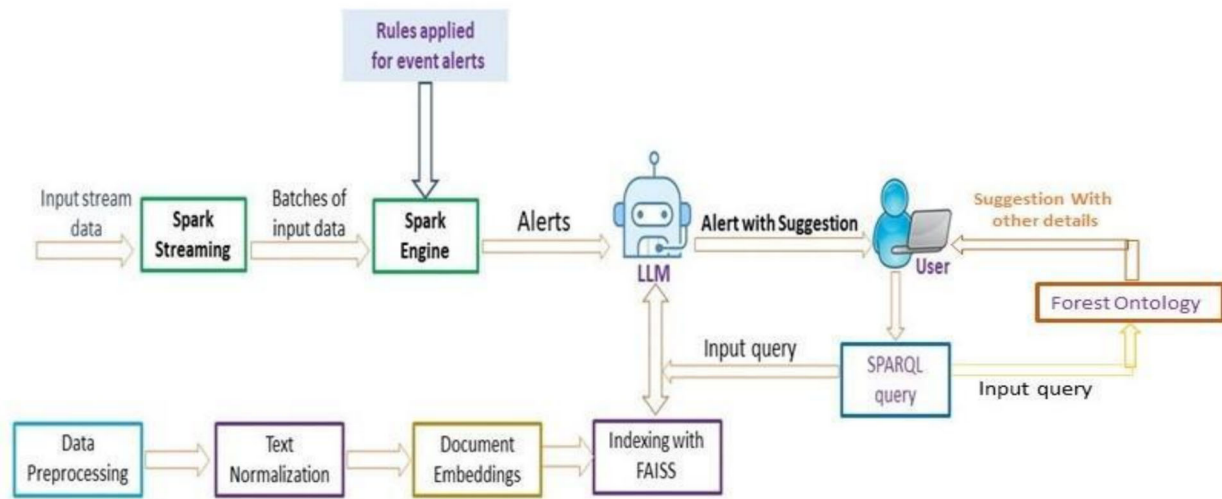


Fig. 9 Integration of spark framework with LLMs and Ontology

4 Result and discussion

This research aims to develop a DSS to help prevent and manage forest fires. In our previous work, we used sensor data from Montesinho Natural Park for its creation. We utilize four types of weather sensors: precipitation, wind speed, humidity, and temperature. Additionally, we incorporate survey reports to supplement the data for our model. Our ontology specifies sensor locations based on these reports. Using SWRL rules, our ontology provides precise recommendations when processed with the reasoner in the Protégé tool [47]. We have integrated 43 rules into our model, which was developed with the help of the SSN ontology. We use SPARQL [48] (SPARQL Protocol and RDF Query Language) queries to obtain comprehensive results, executing 97 queries on this model and producing accurate outcomes [49].

To extend this work, we enhanced the forest fire ontology by incorporating a wide range of forest management system documents from countries such as Australia, Canada, and India, thereby improving the precision and comprehensiveness of the ontology. We adopted a modular and structured integration approach, wherein separate sub-ontologies were developed for each country. These sub-ontologies captured essential elements such as fire danger indices, prevention strategies, alert levels, and emergency response protocols. Official fire management documents, protocols, and guidelines were sourced from certified government websites to ensure the accuracy and consistency of the incorporated knowledge. The sub-ontologies were aligned using standard ontology matching techniques, with equivalent concepts linked via constructs such as `owl:equivalentClass` and `rdfs:subClassOf` to resolve terminological discrepancies for example, mapping “controlled

burn” in Australia to “prescribed burn” in Canada. Conflicts in thresholds, rules, or terminology were identified using reasoning tools such as HermiT [50] in the Protégé environment. To manage regional variations, properties like `hasRegion` and `appliesToCountry` were introduced, enabling the ontology to differentiate rules and concepts based on geographic context. Rule sets were modularized to allow each region’s policies to function independently, and provenance metadata, including source URLs and country references, was attached to each integrated element to ensure traceability and facilitate updates.

In addition to the ontology extension, we introduced Spark Streaming-based alert mechanisms and incorporated LLMs to support automation and reduce human intervention. Numerous forest fire documentation sources were consulted to inform the development of LLM-based components, enabling the models to generate context-aware precautionary suggestions. Over 110 SWRL rules were integrated into the ontology and are actively utilized in the Spark-based streaming alert detection pipeline. When environmental thresholds are breached, Spark triggers real-time alerts, which are sent to the LLMs. The LLMs analyze the contextual data and return situation-specific recommendations, enhancing the relevance and accuracy of the system’s output. These alerts and suggestions are then applied to the ontology, where reasoning mechanisms determine appropriate courses of action. The outcomes inform decision-making processes for forest authorities, enabling timely interventions. The integrated system combines semantic reasoning, real-time data processing, and intelligent natural language interpretation to support a scalable and automated decision support framework for wildland fire management. Evaluation of the system’s performance is detailed in subsequent sections.

4.1 Evaluation of the forest ontology based on metric counts

Based on the ontology metrics count shown in Table 6, we define the inheritance richness, relationship richness, class relation ratio, axiom, class ratio, and inheritance richness, as detailed in Table 7.

Schema metrics: The ontology can also be described using a 5-tuple model, represented as $O = \langle C, Dr, Sc, Re, Ind \rangle$, where C stands for classes, Dr denotes data properties (attributes), Sc indicates subclasses, Re represents relations between classes, and Ind signifies individuals. Metrics can be evaluated based on Attribute Richness, Relationship Richness, Class Richness, and Average Population.

RR measures the depth of connections between concepts within an ontology. It is determined using Eq. 1:

$$RR = |Prop|/|Sub\ class| + |Prop| \quad (1)$$

where $|Prop|$ represents the total number of properties, encompassing both attribute data and object characteristics (class relationships).

AR is calculated by averaging the number of attributes over the entire class, as shown in Eq. 2:

$$AR = |Attribute|/|Class| \quad (2)$$

where $|attribute|$ represents the total number of data attributes.

Table 6 Metrics available in the Forest ontology

Axioms	5897
Logical axioms count	2129
Class count	1007
Data property count	198
Object property count	454
Annotation Property count	53
Object Property Range	281
Object Property Domain	483
Individual count	587
Total individuals count	407

Table 7 Measurements and values associated with ontology

Schema Metrics	Value	Value [49]
Relationship richness	0.8703	0.8093
Class/relation ratio	0.9781	0.7948
Inheritance richness	0.8961	0.8587
Axiom/class ratio	67.2622	42.1742
Attribute richness	0.9019	0.8289

CR indicates the extent of real-world knowledge represented by the ontology. It is calculated using Eq. 3, by dividing the number of classes with instances by the total number of classes:

$$CR = |Class\ with\ instance|/|Class| \quad (3)$$

AP determines the average number of individuals in each class, expressed in Eq. 4:

$$|AP| = |Individual|/|Class| \quad (4)$$

The computed values of different evaluation metrics for the newly designed ontology are mentioned in Table 7.

We also assess the quality score of the ontology based on the represented knowledge, which reflects the quantity and relationships of the domain knowledge. The score for the ontology model (om) can be calculated using the following Eqs. (5).

$$\begin{aligned} \text{Score om} = & [(|Rel| \times |Class| \times 100) \\ & + (|Subclass| + |Rel|) \\ & \times |Prop|]/(|Subclass| + |Rel|) \times |Class| \end{aligned} \quad (5)$$

An ontology score can also be calculated based on the effectiveness of knowledge base (kb) extraction, following Eq. (6).

$$\text{Score kb} = [(Number of Classes \times 100) + (Individual)/|Class|] \quad (6)$$

Table 8 presents the ontology score calculated using the formulas for scores om and kb.

4.2 LLMs based evaluation

To measure the performance of LLMs, we first extract relevant context from the FAISS database based on the input query. The retrieved context is ranked using cosine similarity, which measures the relevance of the context to the query. Cosine similarity values range from -1 to +1, where values closer to +1 indicate higher relevance. After retrieving the context, it is fed into the LLMs, which generate a response based on this information. We then evaluate the LLM's output by calculating Precision, Recall, and F-measure, comparing the generated response against a predefined test response. Precision indicates the proportion of relevant information in the response, Recall measures

Table 8 Forest ontology score

Evaluation parameter	Score	Score [49]
Score om	96.04	89.05
Score kb	95.02	92.03

Table 9 Query-based LLMs score

Query	Cosine Similarity Score(FAISS context)	Score of response		
		Precision	Recall	F-measure
If Ignition Potential is extremely easy and DMC Rules are difficult, what actions should be taken to reduce forest fire?	0.82604045	0.723	0.685	0.7025
If Ignition Potential is extremely easy, DMC Rules is difficult and extensive, DC Rules is difficult and extensive, Rate of Spread is fast and raining is unknown and wind speed is unknown then what measures should be taken?	0.8750995	0.680	0.750	0.7118
If Ignition Potential is moderately easy, DMC Rules is difficult and extensive, DC Rules is difficult and extensive, the Rate of Spread is slow, while Raining is unknown and Wind Speed is unknown, then what measures should be taken?	0.8609843	0.750	0.779	0.7638
The Ignition Potential is moderately easy, DMC Rules is difficult and extensive, DC Rules is difficult and extensive while the Rate of Spread is slow, the raining status and speed of wind is unknown. What measures should be taken?	0.8217561	0.867	0.654	0.7465
The Ignition Potential is moderately easy, DMC Rules is difficult and extensive and DC Rules is difficult and extensive while the Rate of Spread is slow the raining status and speed of wind is unknown what measures should be taken?	0.88260404	0.786	0.890	0.8329

```

question = '''If Ignition Potential is extremely easy, DMC Rules is difficult and extensive,
DC Rules is difficult and extensive, Rate of Spread is fast and raining is unknown and wind speed
is unknown then what measures should be taken?'''

response = llm_chain.invoke(f"{question} answer by understanding this context {context}")

response['text']

' In such a situation where the fire potential is low but the area is vast and extensive with various challenges, the following measures could be taken to help i
n containing the fire and preventing its spread:\n\n1. Develop an Incident Action Plan (IAP): Create an incident action plan that outlines the tactics and techni
ques for fighting the fire. The IAP should consist of a detailed description of the fire's location, size, and behavior, as well as the resources available to fi
ght it.\n2. Establish Communication Channels: Set up reliable communication channels between the incident command, firefighters, and other stakeholders. This wil
l help to disseminate vital information and ensure that everyone is on the same page.\n3. Assign Tasks: Assign tasks to firefighters based on their strengths, sk
ills, and experience. This ensures that each team member has a specific role in containing the fire.\n4. Use Firefighting Techniques: Employ tried-and-true firef
ighting methods such as direct atta...

```

Fig. 10 LLMs based query result

how much relevant information from the context was captured, and the F-measure provides a balance between Precision and Recall. High values for these metrics suggest that the LLM's response is both accurate and contextually relevant, validating its performance.

In Table 9 the first query was shown in Fig. 10 and its results with proper suggestion and precaution measures.

Table 10 presents the outcomes of LLM, which do not incorporate external documentation guidelines; it solely utilizes the dataset and the Monterio Natural Park survey report, both of which were employed in the development of the ontology in prior studies. Derived from the development of these LLMs, we obtain the results of cosine similarity and the LLMs' answers in the form of true or false after querying the LLMs.

These are sample queries used to test the embeddings, and the returned context was evaluated based on cosine similarity. The LLMs' responses were tested against a test response and have proven to be accurate. The embeddings were stored based on various features such as X and Y coordinates, month, day of the week, FFMC, DMC, DC, ISI, temperature, RH, wind speed, rainfall status, and ignition area. Ground truth rules were applied to these values, indicating whether a fire alarm trigger should be sent or not. The retriever then shifts through the embedding database and retrieves data relevant to the query based on cosine similarity. Subsequently, the model answers based on the received context. The results demonstrate the high accuracy of the retrieved context from the embeddings, facilitating effective decision-making by the LLMs.

Table 10 Query-based cosine similarity and LLMs response

Query	Cosine Similarity	LLMs answer
If X: 8, Y: 6, Month: aug, Day: mon, FFMC: 92.3, DMC: 88.9, DC: 495.6, ISI: 8.5, Temperature: 24.1, RH: 27, Wind: 3.1, Rain: 0.0, Area: 0.0, Ignition Potential: extremely easy, DMC Rules: difficult and extensive, DC Rules: difficult and extensive, Rate of Spread: fast, Start Raining: unknown, Wind Speed: unknown, then should Fire Trigger should be sent?	0.879	True
If X: 1, Y: 4, Month: aug, Day: sat, FFMC: 94.4, DMC: 146.0, DC: 614.7, ISI: 11.3, Temperature: 25.6, RH: 42, Wind: 4.0, Rain: 0.0, Area: 0.0, Ignition Potential: extremely easy, DMC Rules: difficult and extensive, DC Rules: difficult and extensive, Rate of Spread: fast, Start Raining: unknown, Wind Speed: unknown, then should Fire Trigger be sent?	0.882	True
If X: 7, Y: 4, Month: aug, Day: sun, FFMC: 81.6, DMC: 56.7, DC: 665.6, ISI: 1.9, Temperature: 21.2, RH: 70, Wind: 6.7, Rain: 0.0, Area: 11.16, Ignition Potential: moderately easy, DMC Rules: difficult and extensive, DC Rules: difficult and extensive, Rate of Spread: slow, Start Raining: unknown, Wind Speed: unknown, then should Fire Trigger should be sent?	0.848	True
If X: 2, Y: 4, Month: aug, Day: sun, FFMC: 81.6, DMC: 56.7, DC: 665.6, ISI: 1.9, Temperature: 21.9, RH: 71, Wind: 5.8, Rain: 0.0, Area: 54.29, Ignition Potential: moderately easy, DMC Rules: difficult and extensive, DC Rules: difficult and extensive, Rate of Spread: slow, Start Raining: unknown, Wind Speed: unknown, then should Fire Trigger be sent?	0.883	True
If X: 4, Y: 3, Month: aug, Day: sun, FFMC: 81.6, DMC: 56.7, DC: 665.6, ISI: 1.9, Temperature: 27.8, RH: 32, Wind: 2.7, Rain: 0.0, Area: 6.44, Ignition Potential: moderately easy, DMC Rules: difficult and extensive, DC Rules: difficult and extensive, Rate of Spread: slow, Start Raining: unknown, Wind Speed: unknown, then should Fire Trigger should be sent?	0.873	True

4.3 SPARQL query performance

Figure 11 depicts the query executed via the LLMs ontology, representing it as an individual component in the proposed work. However, similar queries were executed through combinations of the LLMs ontology and Spark. It's observed that the execution time for Q3 and Q5 varies significantly, with combined models taking more time for the same queries compared to individual models due to complex query. Table 11 illustrates how our model surpasses existing works.

4.4 Comparison with existing literature work

The comparison of different fire detection approaches highlights key strengths and limitations shown in Table 11. Ontology-based models offer structured reasoning and a well-defined knowledge representation but struggle with real-time adaptability, making them less effective for immediate fire response scenarios. Sensor-based IoT systems provide fast alerts, leveraging real-time environmental data; however, they suffer from high false alarm rates

due to sensor noise and environmental variations. Machine Learning (ML)-based approaches demonstrate high accuracy, particularly when trained on large datasets, but they require constant retraining to adapt to different environmental conditions and geographical regions, limiting their generalizability. In contrast, the proposed model (Ontology + LLMs + Spark Streaming) combines the strengths of these traditional methods by offering real-time automation, precautionary suggestions, and better scalability. By integrating knowledge-based reasoning, machine learning, and real-time data streaming, the proposed approach ensures more adaptive, precise, and efficient fire management, addressing the shortcomings of existing systems.

The Table 12 presents a comparative overview of semantic and intelligent systems in forest fire management, highlighting various ontology-driven or hybrid approaches designed to enhance decision-making in dynamic, fire-prone environments. Models like Chandra et al. and Masa et al. rely on structured ontologies and SWRL rules, providing strong domain interpretability but lacking real-time adaptability. On the other hand, Gao et al. introduce

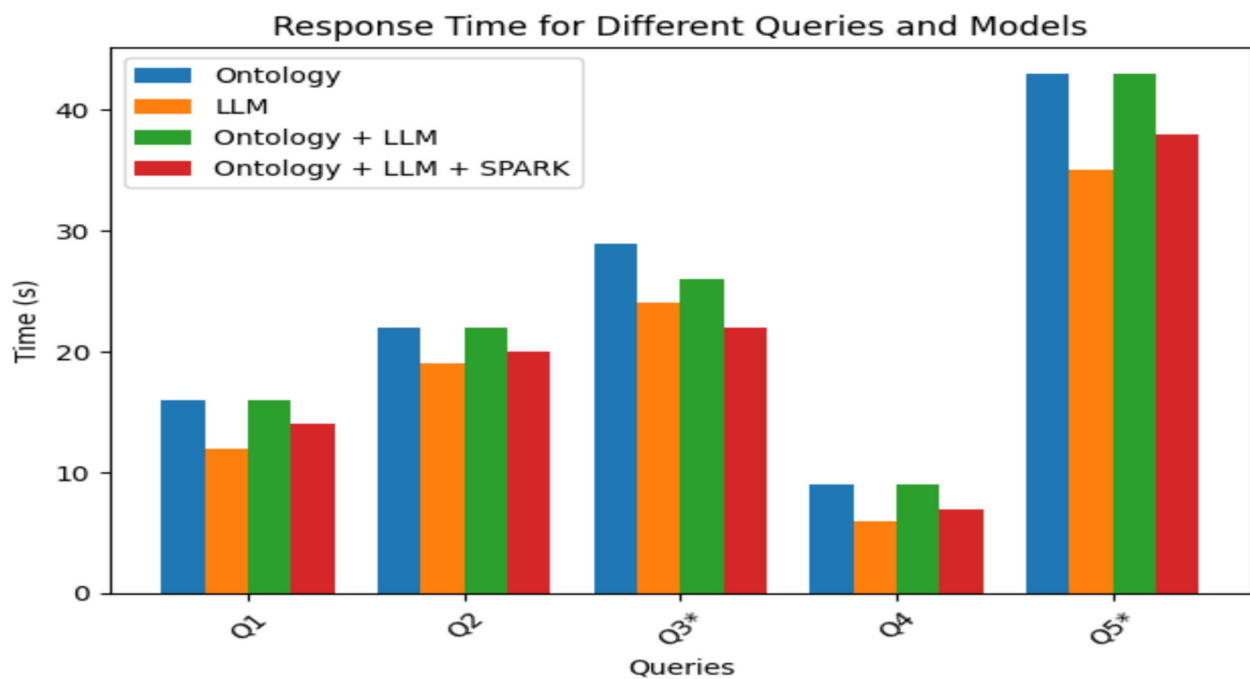


Fig. 11 SPARQL query performance analysis

weather-based predictions, while Ge et al. combine machine learning and spatio-temporal knowledge graphs for real-time fire prediction, though both models face scalability challenges.

The Proposed Model stands out by integrating LLMs with semantic rules and Spark Streaming, enabling both contextual reasoning and real-time alerts. With advanced ontology evaluation metrics and automated response generation, it minimizes manual intervention. Although traditional models excel in static decision support, they lack the ability to handle dynamic conditions. The Proposed Model addresses this gap by offering better scalability, real-time capabilities, and adaptability, making it the most comprehensive solution for fire management.

5 Conclusion and future work

This paper utilizes ontology, LLMs, and Spark streaming to detect forest fires in the forest fire management system. Previous research has primarily focused on ontology and SWRL rules-based forest fire management, which requires

significant human intervention. To address this limitation, the present work combines Spark streaming for real-time data processing and rules-based alert generation, along with LLMs and ontology for automatically generating precautionary suggestions and determining appropriate actions. This integration minimizes human intervention and enables instant action orders from forest officers to ground action teams. To evaluate the performance of the model, we assessed the ontology based on metrics such as schema, query response time, LLMs response score, and cosine similarity. The developed ontology encompasses a broader range of knowledge related to forest management compared to previous versions, as evidenced by various scores. This comprehensive model yields more accurate results. The present model is semi-automated and requires modifications when applied to specific geographic locations, as it relies on local environmental knowledge. This is a limitation of the current work.

Future efforts will focus on developing automated methods to adapt core knowledge components using region-specific data, which will improve the model's scalability and applicability across various fire-prone

Table 11 Comparison of different fire detection approaches

Methodology	Approach	Dataset Used	Evaluation Metrics	Findings
Proposed Model (Ontology + LLMs + Spark Streaming)	Semantic rules, real-time alerts, automated reasoning	Montesinho Fire Dataset	Precision, Recall, F1-score, Cosine Similarity	High accuracy, automated reasoning, minimal human intervention
Ontology-Based DSS [36]	OWL ontology + SWRL Rules	Montesinho Dataset	Query-based accuracy, execution time	High interpretability, but lacks real-time adaptability
Ontology-Based Fire Weather Index [51]	Semantic reasoning over WSN data	Fire Weather Index Sensor Data	Query response time, inference accuracy	Good accuracy but lacks scalability with real-time data
Sensor-Based Fire Detection [52]	IoT and Wireless Multimedia Sensor Networks (WMSN)	IoT sensor data	Accuracy 98.28%	Fast detection, but high false alarms due to environmental noise
WSN + K-means Clustering [14]	Wireless Sensor Networks (WSN) for fire zone prediction	Sensor data from high-risk fire zones	Accuracy 0.987, (Precision, Recall, F1-score = 0.975)	98% accuracy, but high energy consumption
ML-Based Fire Prediction [53]	Random Forest (RF) and SVM models	MODIS Fire Dataset (India)	Accuracy 86%	94% precision, but requires large labeled data
ML-Based Fire Susceptibility Mapping [54]	Extreme Gradient Boosting (XGBoost), RF	Fire Data (Southeast China)	Accuracy 81.59%, (Precision 82.02, Recall 81.03, F1-score 80.47)	RF outperformed Bayesian Model (AUC = 0.97)

Table 12 Comparison with existing work based on methodology and results

Reference	Objective	Methodology	Technology used	Result
Chandra et al. [49]	Forest fire management using semantic sensor technology	SSN ontology and SWRL rules based DSS system	OWL language for ontology and SWRL rules development	Ontology metrics based evaluation, Query based evaluation
Masa et al. [55]	A framework based on ontology for representing and interlinking wildfire events, utilized for decision support in forest fire management	ONTO-SAFE ontology	Ontological web language and semantic reasoning	Incorporating expert knowledge in the form of constraints and rules to identify patterns and situations within domain knowledge, aiding end-users in advanced decision-making
Gao et al. [51]	Integration of semantic reasoning and ontology to design semantic fire weather index	Rule based inference algorithm and Inverse Distance Weighting (IDW)	Wireless sensor network and Semantic reasoning	Queries used to evaluate the models efficiently in terms of accuracy and precision
Ge et al. [56]	Forest fire analysis using the semantic rules of the knowledge graph	Spatio-temporal knowledge graph	Machine learning and rule based reasoning	Real time fire prediction using precision, recall and F1 score
Proposed model	Forest Fire Management	Forest ontology, SWRL rules, Spark streaming based alert, and LLMs based precautionary suggestion	OWL web language, LLMs, Spark	Cosine score, Precision, recall, F1 score, Ontology based metrics evaluation. Attribute Richness, Class Richness, Relationship Richness, and Average Population

ecosystems. To enhance efficiency and reliability, additional rules and machine learning techniques will be incorporated. These improvements will support more accurate and responsive decision-making across different forest fire scenarios and datasets.

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Declaration

Conflict of interests The authors declare no competing interests.

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